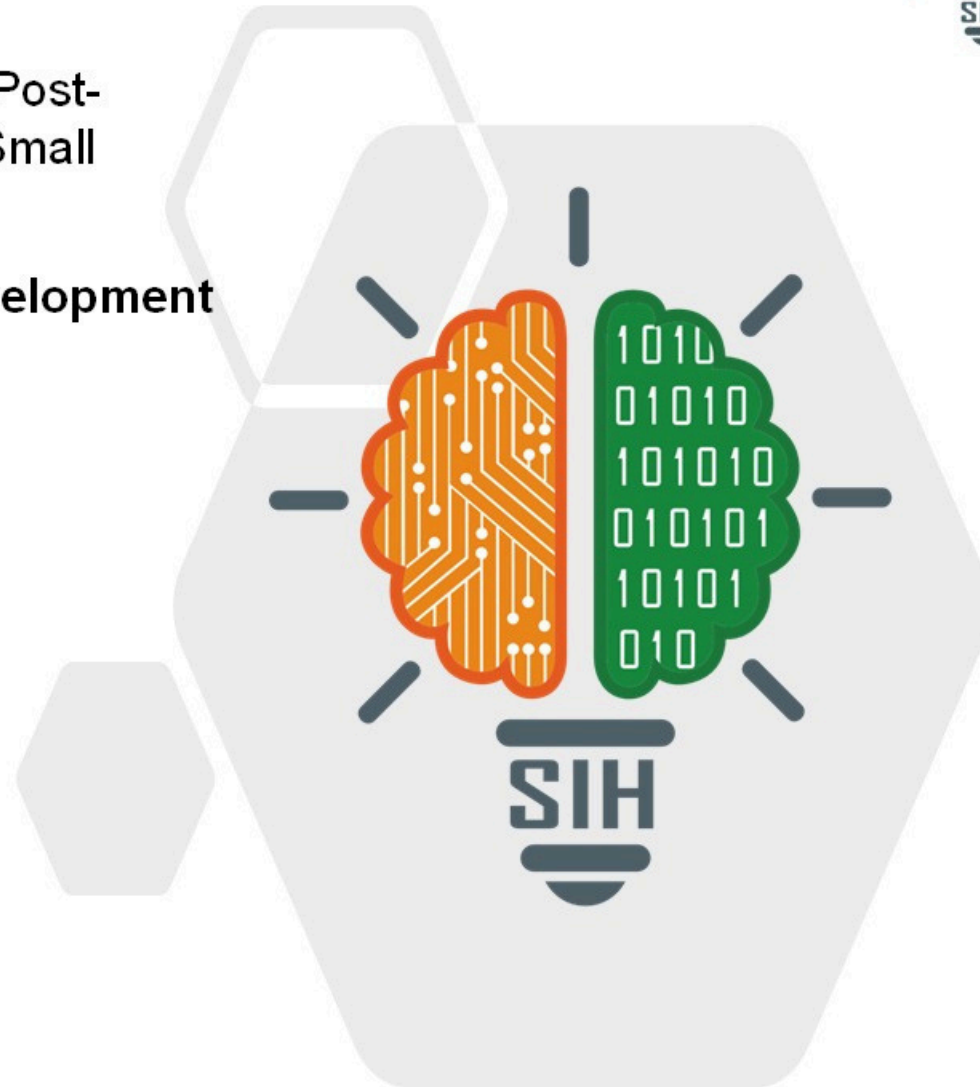


SMART INDIA HACKATHON 2026



- **Problem Statement Title-** FarmFlow: AI-Driven Post-Harvest Decision & Market Access Platform for Small Farmers
- **Theme-** Agriculture, Food Tech and Rural Development
- **PS Category-** Software
- **Team Name-** Bug Slayers
- **Team Members-**
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 - Harshit Singh

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THE PROBLEM

- Small lots (50-200kg) don't meet a cold store's bulk minimum
- No affordable storage means selling immediately, at whatever price
- Result: ₹92,000 Cr lost every year, hitting the 82% who farm under 1 hectare hardest



₹92,000 Cr Estimated annual value of post-harvest losses

82% Farmers operate on less than 1 hectare

40-50%+ Logistics can consume a major share of farm-gate value

The gap FarmFlow is designed to solve.

THE SOLUTION: FARMFLOW

- Lot aggregation - pool small lots into one booking
- Shared storage booking and shared pickup
- AI sell-vs-store call, direct market access



The problem isn't producing food. It's keeping its value after harvest.

FARMFLOW – SYSTEM ARCHITECTURE (3-LAYER)

1. CLIENT / EXPERIENCE LAYER

Farmer App

Farmer App

- React, Vite
- Tailwind CSS
- React Router

Voice Booking

- Web Speech API
- Hindi / Punjabi / English
- Intent Parser & Extractor

Freshness Check

- MobileNetV3-small
- ONNX Runtime Web

See full flow →

Offline Support

- IndexedDB / Dexie
- Local Queue
- Auto Sync when Online

2. APPLICATION / INTELLIGENCE LAYER

API & Backend

- Python, FastAPI
- Uvicorn, Pydantic
- SQLAlchemy

Sell / Store Decision Engine

- Method: compares price forecast, freshness & storage cost →
- recommends Sell Today or Store

Storage Matching & Aggregation

- Matches nearby storage & pools
- farmer lots for shared transport

Trust & Receipts

- QR digital receipts
- Ed25519 Signature, HMAC-SHA256
- Verifiable offline

3. DATA / PERSISTENCE LAYER

Database

- SQLite
- Bookings, farmer, storage and receipt data

Local Data (Offline)

- IndexedDB / Dexie
- Offline bookings, sync queue, cached app data

AI Model Assets

- ONNX Model Files
- Runs locally in the browser

FRESHNESS CHECK

HOW IT WORKS

- 1 Photo / Video**
Farmer captures a quick photo or clip in the app
- 2 Quality Check**
Checks the image is clear enough to analyze
- 3 AI Analysis**
MobileNetV3 model scores freshness
- 4 Result**
Freshness grade & recommendation shown

- Quality Grade (A / B / C)
- Defect Detection
- Spoilage Risk (Low / Med / High)
- Shelf-life Estimate

FEASIBILITY

- Uses existing cold-storage infrastructure
- Aggregates small farmer lots into viable bookings
- Uses existing transport through shared pickup
- Offline-first farmer app supports low-connectivity areas
- AI models assist sell-vs-store & freshness decisions

VIABILITY

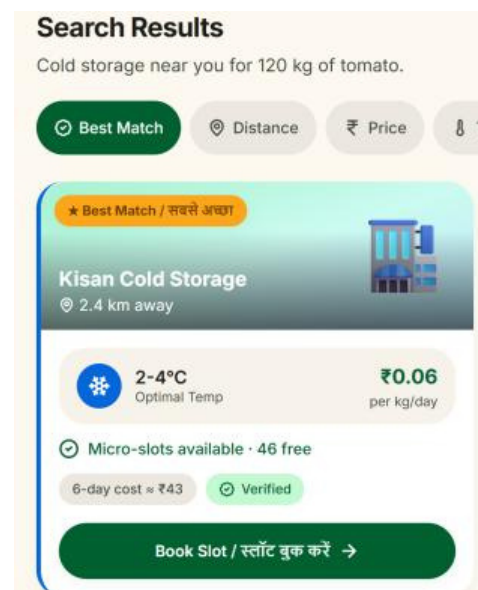
- Revenue: booking + logistics + buyer transaction fees
- Low asset model, no need to build new cold stores
- Starts with one community, then district, state, Pan-India
- Benefits farmers, storage owners, transporters and buyers

POTENTIAL RISKS & MITIGATION

- **Risk:** Existing cold stores may hesitate to accept fragmented small-farmer bookings
- **Solution:** Aggregate demand and automate slot utilisation to create value for storage owners.
- **Risk:** Scaling across multiple districts may make coordination complex
- **Solution:** Use hub-based aggregation with automated farmer storage-buyer matching.

PROVEN IN THE PROTOTYPE

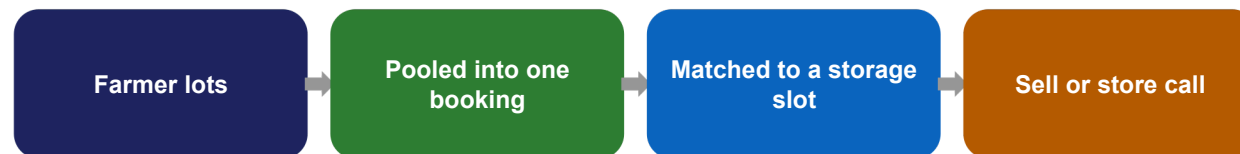
- 225 automated checks pass, including a 20-thread booking race test
- Vision model measured at 98.9% crop, 96.6% ripeness accuracy on unseen photos
- Runs on an ordinary Android phone, on free-tier hosting at pilot scale



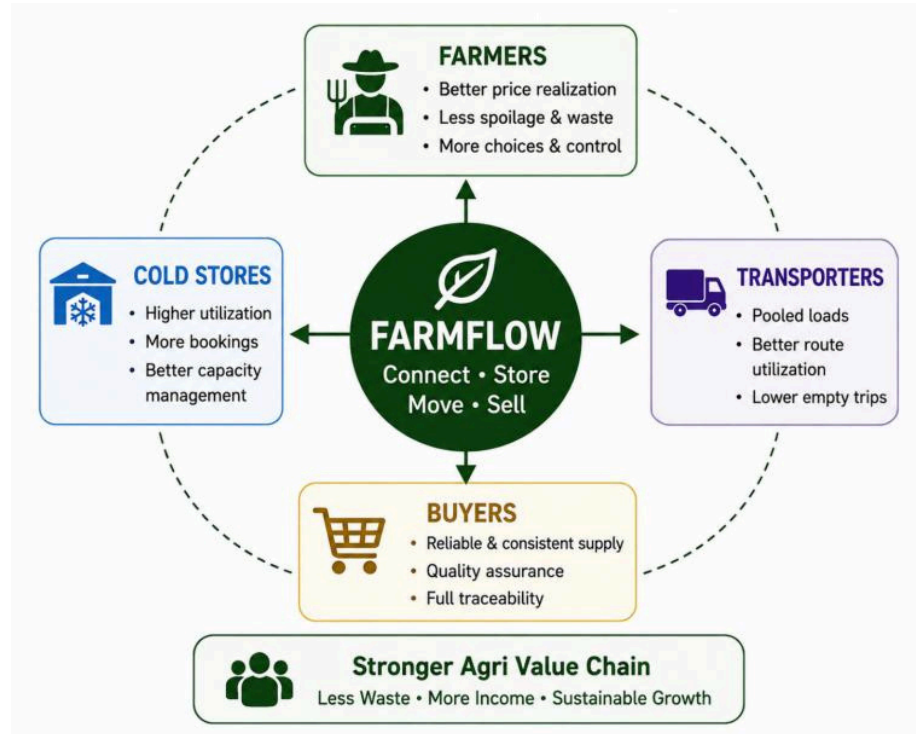
Matching an existing cold store



Pooling nearby farmers' lots



No new cold stores built. Existing capacity, made reachable.



BENEFITS

- Affordable access to storage & transport
- Lower transaction costs through aggregation
- Better sell-vs-store decisions

Market Landscape study

WHAT EXISTS TODAY	FARMFLOW
 Large / organized supply chains	✓ 50–500 kg small farmers
 Separate storage booking	✓ Shared storage through aggregation
 Individual transport	✓ One pickup for nearby farmers
 Basic market access	✓ AI Sell vs Store decision
 Standard digital apps	✓ Voice + Hindi/Punjabi/Regional languages
 Wholesale connections	✓ Direct B2B buyer marketplace

IMPACT

- Reduced post-harvest losses and forced selling
- Higher utilisation of existing cold-chain capacity • Stronger farmer-to-market connectivity
- Creates a stronger post-harvest value chain benefiting all key stakeholders — Farmers, Cold-Storage Owners, Transporters, Buyers, and Consumers.

Existing players referenced

- [DeHaat](#)—farmerservices & market linkage
- [Ninjacart](#)—farm-to-market supply chain
- [WayCool](#)—B2B food supply chain

Each addresses one part of the post-harvest journey; FarmFlow connects AI decision-making, aggregation, shared storage and direct buyer access in one flow.

Datasets, standards and methods used in this build

- [VegNetvegetable-quality dataset](#)—Ahmad et al., Mendeley Data, CC BY 4.0
- [Fruit and Vegetable Disease dataset](#)—Kaggle
- [Searching for MobileNetV3](#)—Howard et al., ICCV 2019

Aligned government priorities

- [MoFPI Cold Chain Scheme](#)—cold-chain access
- [e-NAM](#)—market & price discovery
- [SDG 2](#)—Zero Hunger
- [SDG 8](#)—Decent Work
- [SDG 12](#)—Responsible Consumption
- [SDG 13](#)—Climate Action

STANDARDS & RESEARCH SOURCES

- [WDRA electronic Negotiable Warehouse Receipt framework](#)
- [Agmarknet](#)—daily mandi prices
- [NABCONS post-harvest loss study for MoFPI](#)—54 crops, 2020-22
- [RFC 8032](#)—Edwards-Curve Digital Signature Algorithm